$$\dot{V} \le -\nu \|\underline{x}\|_2^2 + 2\gamma_p \|\underline{x}\| \|\underline{r}\|$$

We know that $\underline{r}(\underline{x}(t), t)$ is 2nd order in $\underline{x}$, which means

$$\lim_{\|\underline{x}\| \to 0} \frac{\|\underline{r}(\underline{x}, t)\|}{\|\underline{x}\|} = 0 \quad \forall t$$

More precisely, for any $\varepsilon > 0$, $\exists \delta(\varepsilon) > 0$ so that

$$\frac{\|\underline{r}(\underline{x}, t)\|}{\|\underline{x}\|} \le \varepsilon \quad \text{for all } \|\underline{x}\| \le \delta \text{ and all } t.$$

Equivalently, for all $\|\underline{x}\| < \delta$ we have

$$\|\underline{r}(\underline{x}, t)\| \le \varepsilon \|\underline{x}\| \quad \text{for all } t$$

Thus, for all $\underline{x}$ with $\|\underline{x}\| \le \delta$

$$\dot{V} \le -\nu \|\underline{x}\|_2^2 + 2\gamma_p \varepsilon \|\underline{x}\|_2^2$$

Now, choose ε sufficiently small that

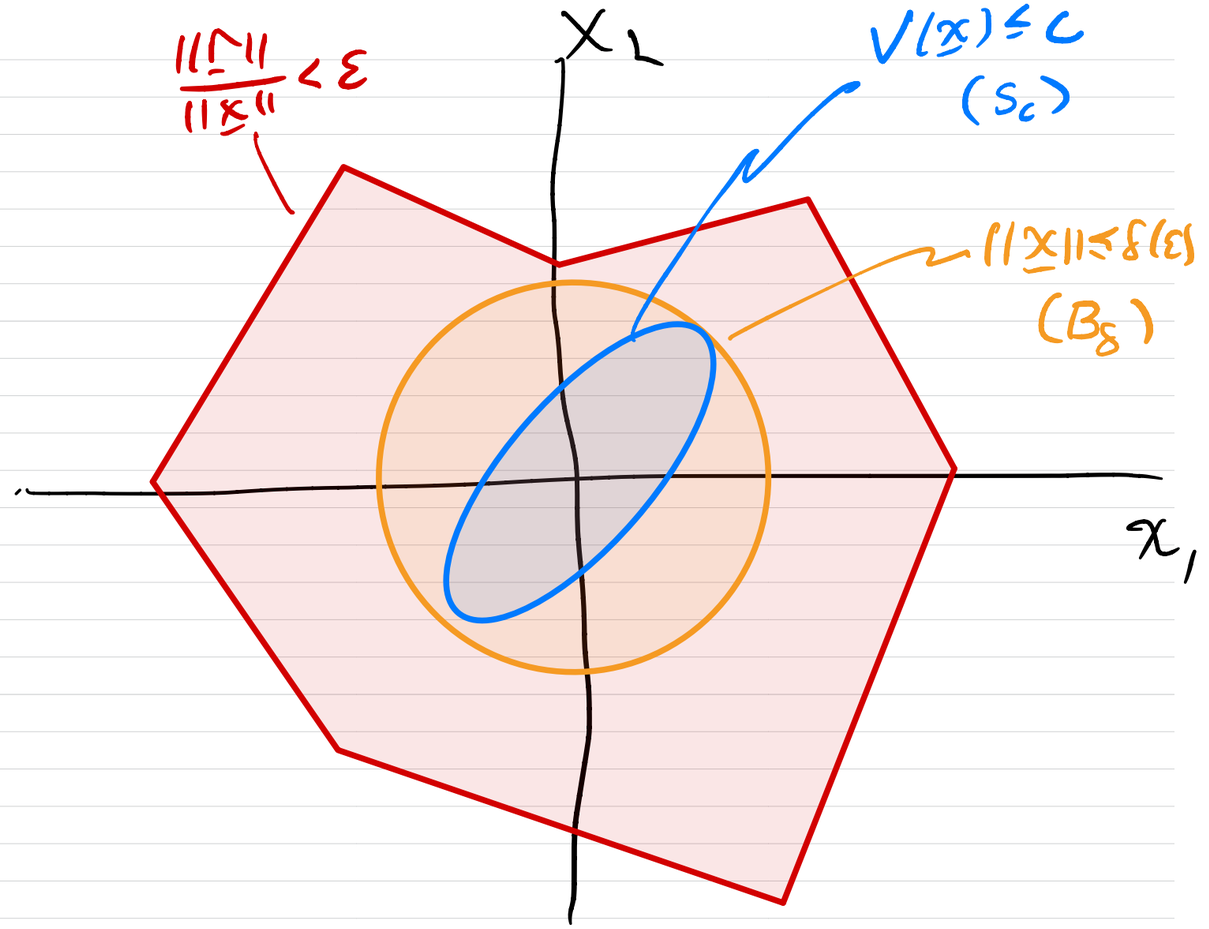
$$\nu_1 = \nu - 2\gamma_p \varepsilon > 0 \implies \varepsilon < \frac{\nu}{2\gamma_p}$$

Then

$$\dot{V} \le -\nu_1 \|\underline{x}\|_2^2 \quad \text{for all } \underline{x} \text{ with } \|\underline{x}\| < \delta$$

$\nu = \min_t \lambda_{\min}(Q(t)) > 0$

$\gamma_p = \max_t \|P(t)\|_i$



• Let c be such that the contour $V(\underline{x}) = c$ is entirely contained in the ball (B_δ) $\|\underline{x}\| \le \delta$.

• Let S_c be the set (ellipsoid) $S_c = \{\underline{x} \mid V(\underline{x}) \le c\} \subset B_\delta$

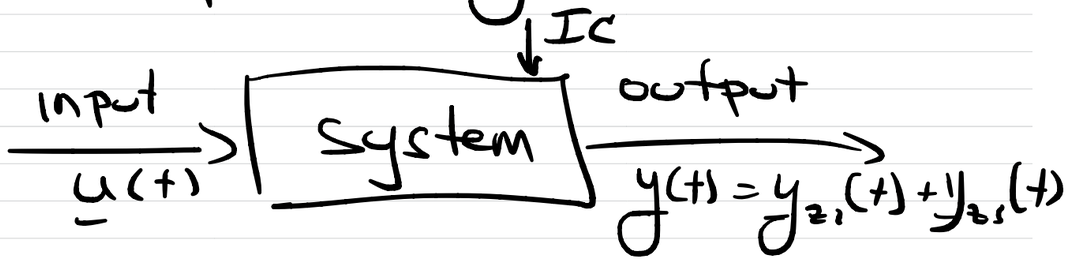
- Then for any $\underline{x} \in S_c$, $\dot{V} \leq -\alpha_1 \|\underline{x}\|_2^2 < 0$
  Hence S_c is an *invariant set*
  i.e. $\underline{x}_0 \in S_c \Rightarrow \underline{x}(t) \in S_c \ \forall t \geq t_0$
- Since $S_c \subset B_\delta$, we then have that if $\underline{x}_0 \in S_c$, $\dot{V}(\underline{x}(t)) \leq -\alpha_1 \|\underline{x}(t)\|_2^2$ for all $t \geq t_0$, and thus $\underline{x}(t)$ converges exponentially to 0 from any such initial condition
- This is *local* stability, and S_c is an estimate of the region ("basin") of attraction.

A Different Notion of stability:

"input-output" stability

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Focused on the nature of the *forced* response, as opposed to the nature of the homogeneous response.



I/O stability means there exists $\gamma < \infty$ such that, for every $\underline{u}(t)$

$$\max_t \|y_{zs}(t)\| \leq \gamma \max_t \|\underline{u}(t)\|$$

Example: SISO

$$y_{zs}(t) = \int_{t_0}^{t} g(t,\tau) u(\tau) \, d\tau$$

$$y_{zs}(t) = \int_{t_0}^{t} g(t,\tau)\,u(\tau)\,d\tau$$

so

$$|y_{zs}(t)| \le \int_{t_0}^{t} |g(t,\tau)|\,|u(\tau)|\,d\tau$$

$$|y_{zs}(t)| \le \max_{[t_0,t]} |u(\tau)| \int_{t_0}^{t} |g(t,\tau)|\,d\tau$$

$$\Rightarrow \max_t |y_{zs}(t)| \le \max_t |u(t)| \int_{t_0}^{\infty} |g(t,\tau)|\,d\tau$$

So we need

$$\gamma = \int_{t_0}^{\infty} |g(t,\tau)|\,d\tau < \infty$$

i.e. impulse response $g(t,\tau)$ must be "absolutely integrable"

The equivalent MIMO condition is that

$\|G(t,\tau)\|_i$ is integrable,

where G is the $p \times m$ impulse response matrix

But this is equivalent to requiring each of the scalar entries $g_{ij}(t,\tau)$ to be absolutely integrable.

=

Are Lyap. stability + IIO stability the same?

Not always!

Consider $\dot{x} = u$

$y = x$

$x(t) = x_0 \ \forall t$

Homog eq'n $\dot{x} = 0$ is isL stable

But with $u(t) = 1$ for all $t \ge 0$

$y(t) = x_0 + t \Rightarrow$ Not IIO stab

More generally, recall

$$y_{zs}(t) = D(t)\underline{u}(t) + \int_{t_0}^{t} C(t)\Phi(t,\tau)B(\tau)\underline{u}(\tau)d\tau$$

Hence

$$\max_t \|y_{zs}(t)\| \le \max_t \|\underline{u}(t)\| \left[\|D(t)\|_i + \int_{t_0}^{\infty} \|C(t)\|_i \cdot \|\Phi(t,\tau)\|_i \|B(\tau)\|_i \, d\tau \right]$$

Suppose B, C, D have finite induced norms for all t, i.e.

$$\|B(t)\|_i \le \gamma_B, \ \|C(t)\|_i \le \gamma_C, \ \|D(t)\|_i \le \gamma_D$$

Then

$$\max_t \|y_{zs}(t)\| \le \max_t \|\underline{u}(t)\| \left[\gamma_D + \gamma_B \gamma_C \int_{t_0}^{\infty} \|\Phi(t,\tau)\|_i d\tau \right]$$

So $\gamma < \infty$ (I/o stable) if $\|\Phi\|_i$ is integrable

Note: Exp stable $\Rightarrow \|\Phi(t,\tau)\|_i \le \gamma_x e^{-\lambda(t-\tau)}$, $\lambda > 0$

and $\int_{t_0}^{\infty} e^{-\lambda(t-\tau)} d\tau < \infty$ for all finite t $\Rightarrow$ Exp stability $\Rightarrow$ I/O stability

Is the opposite true? i.e. does I/O stability => exp. stability?

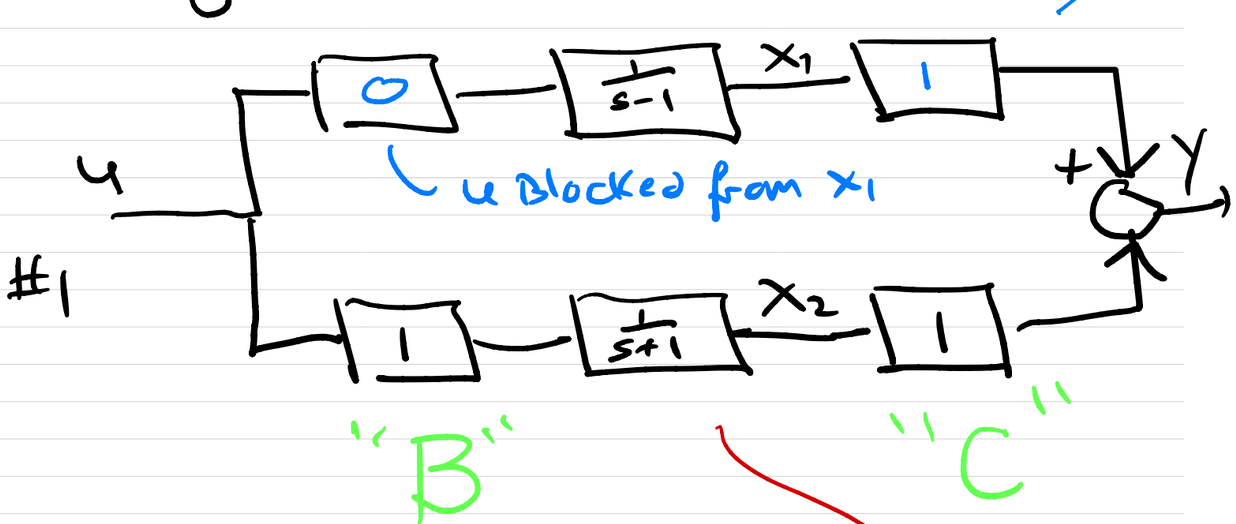
Maybe:

Consider two systems

#1 $\dot{x}_1 = x_1$ $\quad$ $\dot{x}_1 = x_1 + u$

$\dot{x}_2 = -x_2 + u$ $\quad$ $\dot{x}_2 = -x_2 + u$

$y = x_1 + x_2$ $\quad$ $y = x_2$

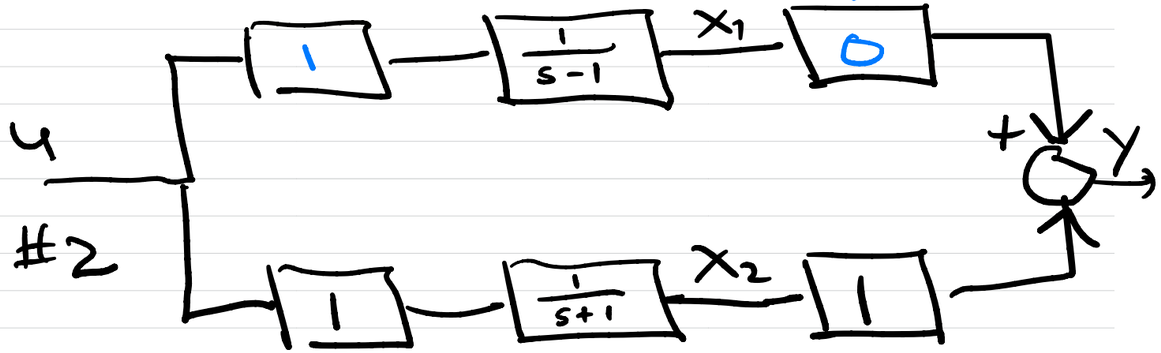


Clearly there is a more sensitive dependence on nature of B/C here. Can unstable internal effects be "blocked" from y(t) somehow?

- "controllability"
- "observability"

Unstable path } Not stable i.s.L

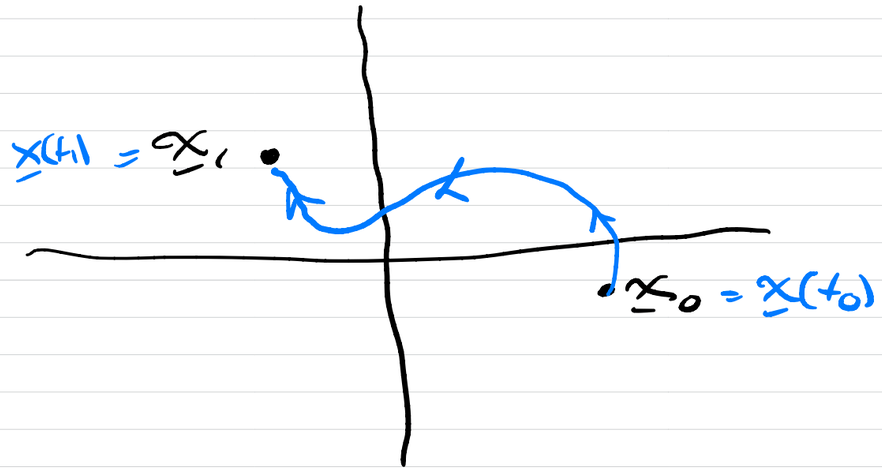
x1 Blocked from y



Stable path // y shows only effect of input on stable path.

Controllability

A system is controllable at t_0 if there exists finite $t_1 > t_0$ such that for any $\underline{x}_0 = \underline{x}(t_0)$ and any $\underline{x}_1 \in \mathbb{R}^n$, there exists $\underline{u}(\tau)$, $t_0 \le \tau \le t_1$, so that $\underline{x}(t_1) = \underline{x}_1$



for any pair $(\underline{x}_0, \underline{x}_1)$ find an input $\underline{u}(t)$ defined over a finite interval $[t_0, t_1]$ which will "steer" the system from $\underline{x}(t_0) = \underline{x}_0$ to $\underline{x}(t_1) = \underline{x}_1$

Note: No assumption that the state can be held at $\underline{x}_1$: i.e. $\underline{x}(t) = \underline{x}_1$, $t \geq t_1$. This is a separate condition, not part of defn of controllability!

When can this be accomplished?

Look at general soln

$$\underline{x}(t) = \Phi(t, t_0)\underline{x}_0 + \int_{t_0}^{t} \Phi(t, \tau) B(\tau) \underline{u}(\tau) \, d\tau$$

we need, for some finite t_1:

$$\underline{x}(t_1) = \underline{x}_1 = \Phi(t_1, t_0)\underline{x}_0 + \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) \underline{u}(\tau) \, d\tau$$

Equiv: (arbitrary vector in $\mathbb{R}^n$)

$$\underline{x}_1 - \Phi(t_1, t_0)\underline{x}_0 = \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) \underline{u}(\tau) \, d\tau$$

must hold for any specified $\underline{x}_0, \underline{x}_1$,

Hence we need

$$\langle \text{any vector in } \mathbb{R}^n \rangle = \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) \underline{u}(\tau) \, d\tau$$

Note $B(\tau)\underline{u}(\tau) = \sum_{j=1}^{m} \underline{b}_j(\tau) u_j(\tau)$, hence

$$\langle \text{any vec in } \mathbb{R}^n \rangle = \sum_{j=1}^{m} \int_{t_0}^{t_1} \underbrace{\Phi(t_1, \tau)\underline{b}_j(\tau)}_{\underline{h}_j(t_1, \tau)} u_j(\tau) \, d\tau$$

$$= \sum_{j=1}^{m} \int_{t_0}^{t_1} \underline{h}_j(t_1, \tau) u_j(\tau) \, d\tau$$

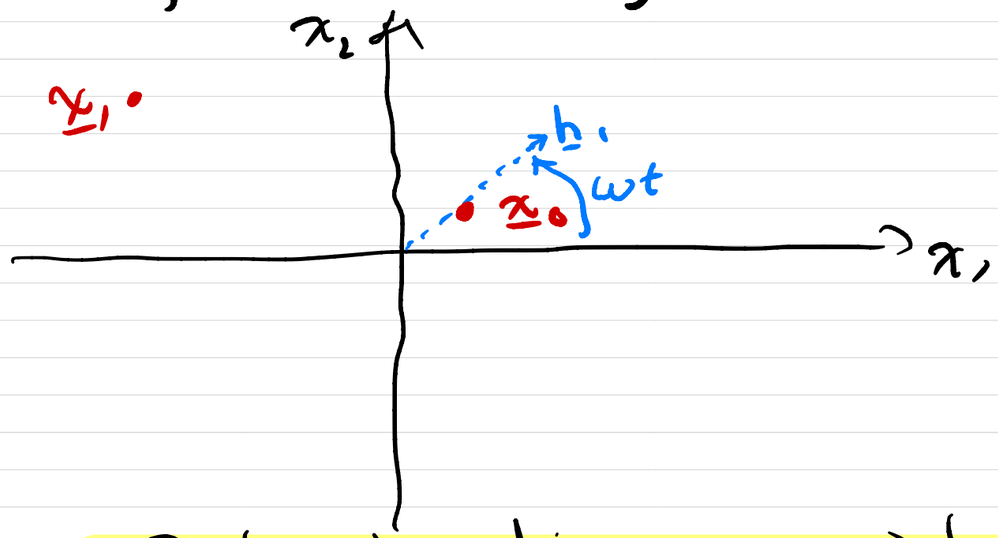
$\Leftarrow$

We need a weighted sum of these m n-dim vectors to add to an arbitrary vector in $\mathbb{R}^n$.

$\underline{h}_j(t_1, \tau) \in \mathbb{R}^n$

$\sum_{j=1}^{m}$ ← Adding together m such vectors.

Typically $m < n$ (usually $m \ll n$) How can the $m \ll n$ vectors $\underline{h}_j(t_1,\tau)$ in any sense produce a basis for $\mathbb{R}^n$?



We need the directions spanned by the time vars in $\underline{h}_j$ (really their integrals) to encompass all of $\mathbb{R}^n$.

How can we tell when the collection $\{\underline{h}_j(t_1,\tau)\}_{j=1}^m$ has this property??

Let

$H(t_1,\tau) = [\underline{h}_1(t_1,\tau)\ \underline{h}_2(t_1,\tau) \cdots \underline{h}_m(t_1,\tau)]$ — $n \times m$ → not square

$= \underline{\Phi}(t_1,\tau)B(\tau)$

Thm:

If there exists finite $t_1 > t_0$ such that

$W_c(t_0,t_1) = \int_{t_0}^{t_1} H(t_1,\tau)H(t_1,\tau)^T d\tau$ — $n \times n$

is positive definite, then system is controllable

controllability gramian

alternately:

$W_c(t_0,t_1) = \int_{t_0}^{t_1} \Phi(t_1,\tau)B(\tau)B^T(\tau)\Phi^T(t_1,\tau)d\tau$

$=$

Note: W_c is at least positive semi-definite, i.e. for any constant $\underline{x} \in \mathbb{R}^n$

$\underline{x}^T W_c(t_0,t_1)\underline{x} \geq 0$

Let $\underline{z}(t_1,\tau) = B^T(\tau)\,\Phi^T(t_1,\tau)\,\underline{x}$

$$\underline{z}^T(t_1,\tau) = \underline{x}^T\Phi(t_1,\tau)B(\tau)$$

and

$$\underline{x}^T W_c(t_0,t_1)\underline{x} = \int_{t_0}^{t_1} \underline{z}^T(t_1,\tau)\,\underline{z}(t_1,\tau)\,d\tau$$

$$= \int_{t_0}^{t_1} \|\underline{z}(t_1,\tau)\|_2^2\, d\tau$$

$$\geq 0$$

Controllability requires $W_c(t_0,t_1)$ to be positive definite, i.e

$$\underline{x}^T W_c(t_0,t_1)\underline{x} > 0 \quad \forall \underline{x} \neq \underline{0}$$

$$\Rightarrow \lambda_{min}(W_c(t_0,t_1)) > 0$$

$$\Rightarrow W_c(t_0,t_1) \text{ is nonsingular}$$

Pf: (By construction!)

Let

$$\underline{u}(t) = \underbrace{B^T(t)\Phi^T(t_1,t)}_{\text{2 changing with time}}\underbrace{W_c^{-1}(t_0,t_1)\left[\underline{x}_1 - \Phi(t_1,t_0)\underline{x}_0\right]}_{\text{constant}}$$

(of course, defined only if $W_c(t_0,t_1)$ is nonsingular/positive definite)

This input will "steer" state from $\underline{x}(t_0) = \underline{x}_0$ to $\underline{x}(t_1) = \underline{x}_1$

(see next page)

Why does this work?

if $\underline{u}(t) = \underline{B}^T(t)\underline{\Phi}^T(t_1,t)\,\omega_c^{-1}(t_0,t_1)\left[\underline{x}_1 - \Phi(t_1,t_0)\underline{x}_0\right]$

constant

2 changing with time

Then

$$\underline{x}(t_1) = \Phi(t_1,t_0)\underline{x}_0 + \int_{t_0}^{t_1} \Phi(t_1,\tau)B(\tau)\underline{u}(\tau)\,d\tau$$

$\underline{u}(\tau) = B^T(\tau)\Phi^T(t_1,\tau)\,\omega_c^{-1}(t_0,t_1)\left[\underline{x}_1 - \Phi(t_1,t_0)\underline{x}_0\right]$ — constant

so

$$\underline{x}(t_1) = \Phi(t_1,t_0)\underline{x}_0 + \left\{\int_{t_0}^{t_1} \Phi(t_1,\tau)B(\tau)B^T(\tau)\Phi^T(t_1,\tau)\,d\tau\right\}\omega_c^{-1}(t_0,t_1)\left[\underline{x}_1 - \Phi(t_1,t_0)\underline{x}_0\right]$$

$\tau\omega_c(t_0,t_1)$ by def'n

$$= \underline{x}_1 \quad \checkmark$$

Note: If we differentiate the expression for $\omega_c(t_0,t)$, we get:

Let $\omega_c(t) \triangleq \omega_c(t_0,t)$

Then
$$\begin{cases} \dot{\omega}_c(t) = A(t)\omega_c(t) + \omega_c(t)A^T(t) + B(t)B^T(t) \\ \omega_c(t_0) = 0 \quad (n\times n \text{ matrix}) \end{cases}$$

Some add'l qualifications of controllability

The system is uniformly controllable if $\exists T < \infty$ such that

$W_c(t, t+T)$ pos def for all t

The system is differentially controllable if $W_c(t_0, t_1)$ is pos def for any $t_1 > t_0$ (i.e $t_1 = t_0 + dt$ for arbitrarily small dt)

How to determine controllability w/o constructing the STM (or solving the DE for $W_c(t)$)?

i.e. tests on $A(t)$ and $B(t)$ directly

Thm:

Suppose $A(t)$ and $B(t)$ are $n-1$ times continuously differentiable:

Let

$$M(t) = \begin{bmatrix} K_0(t) & K_1(t) & \cdots & K_{n-1}(t) \end{bmatrix}$$

where

$$K_0(t) = B(t)$$

$$K_j(t) = \dot{K}_{j-1}(t) - A(t) K_{j-1}(t)$$

$$j = 1, \ldots, n-1$$

Note $K_j(t)$ are $n \times m$ (same size as B) and $M(t)$ is $n \times (n \cdot m)$

If there exists finite t_1 such that $\operatorname{rank}(M(t_1)) = n$

Then system is controllable.

i.e. among the n·m columns of $M(t_1)$, n of them are linearly indep

(Note: for a single input ($m=1$) system, we need $M(t_1)$ to be nonsingular.)

For LTI, the above condition becomes:

Let

$$M_c = [B \; -(AB) \; A^2B \; \cdots \; (-1)^{n-1}A^{n-1}B]$$

and system is controllable if

$$\text{rank}(M_c) = n$$

Equivalently

$$M_c = [B \; AB \; A^2B \; \cdots \; A^{n-1}B]$$

(since changing signs on columns doesn't change rank calculation)

LTI controllability matrix.

Note: if LTI system is controllable, it is both uniformly and differentially controllable

$\Leftrightarrow$ (LTI) This test is equivalent to saying

$$\text{Span}\{A^k b_j\} = \mathbb{R}^n \quad k = 0, \ldots, n-1, \quad j = 1, \ldots, m$$

We could have deduced this LTI condition directly w/o going through the general gramian test etc. How?

for LTI

$$\underline{x}(t_1) = e^{A(t_1-t_0)}\underline{x}_0 + \int_{t_0}^{t_1} e^{A(t_1-\tau)} B\,\underline{u}(\tau)\,d\tau$$

where

$$B\,\underline{u}(\tau) = \sum_{j=1}^{m} \underline{b}_j u_j(\tau)$$

and recall that Caley-Hamilton allows us to express e^{At} as a finite sum

$$e^{At} = \sum_{k=0}^{n-1} A^k \alpha_k(t)$$

$\Rightarrow$

$$\underline{x}(t_1) - e^{A(t_1-t_0)}\underline{x}_0 = \sum_{k=0}^{n-1}\sum_{j=1}^{m} A^k \underline{b}_j \int_{t_0}^{t_1} \alpha_k(t_1-\tau)\,u_j(\tau)\,d\tau$$

$\underline{x}(t_1) - e^{A(t_1-t_0)}\underline{x}_0$: arb vector in $\mathbb{R}^n$

$\sum_{k=0}^{n-1}\sum_{j=1}^{m} A^k \underline{b}_j \int_{t_0}^{t_1} \alpha_k(t_1-\tau)\,u_j(\tau)\,d\tau$: linear comb of $\{A^k \underline{b}_j\}_{\substack{k=0,\dots,n-1 \\ j=1,\dots,m}}$

Hence if $\text{span}\{A^k \underline{b}_j\} = \mathbb{R}^n$

then system is controllable